

Chapter 08 Rotational Equilibrium and Rotational Dynamics

Torque

τ , torque, is the tendency of a force to rotate an object about some axis.

$$\vec{\tau} = \vec{r} \times \vec{F}$$

where $\vec{r}$ is the position vector, $\vec{F}$ is the force.

The direction of τ obeys right hand rule.

Equilibrium

To ensure mechanical equilibrium, we should ensure both translational equilibrium and rotational equilibrium.

- Translation equilibrium:

$$\sum \vec{F} = 0$$

- Rotational Equilibrium:

$$\sum \vec{\tau} = 0$$

Center of Gravity

The coordinates of the center of gravity can be found from the sum of the torque acting on the individual particles being set equal to the torque produced by the weight of the object.

The center of gravity of a homogenous, symmetric body must lie on the axis of symmetry. And the center of gravity of such an object is the geometric center of the object.

Torque and Angular Acceleration

The angular acceleration is directly proportional to the net torque.

$$\sum \tau = I\alpha$$

Which is similar to $\sum F = ma$

Thus it is Newton's second Law for a rotating object.

Moment of Inertia

The mass analog is called the **moment of inertia**, I .

$$I = \sum mr^2$$

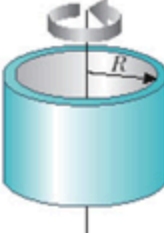
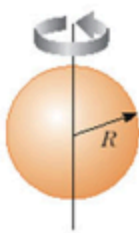
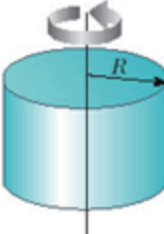
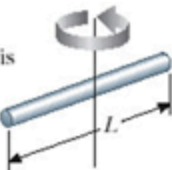
or

$$I = \int r^2 dm$$

Differences between moment of inertia and mass: The moment of inertia depends on the quantity of matter and its **distribution** in the rigid object, and also depends on the **location of the axis of rotation**.

TABLE 8.1

Moments of Inertia for Various Rigid Objects of Uniform Composition

Hoop or thin cylindrical shell $I = MR^2$ 	Solid sphere $I = \frac{2}{5} MR^2$ 
Solid cylinder or disk $I = \frac{1}{2} MR^2$ 	Thin spherical shell $I = \frac{2}{3} MR^2$ 
Long thin rod with rotation axis through center $I = \frac{1}{12} ML^2$ 	Long thin rod with rotation axis through end $I = \frac{1}{3} ML^2$ 

Rotational Kinetic Energy

An object rotating about some axis with an angular speed ω has rotational kinetic energy

$$KE_r = \frac{1}{2} I \omega^2$$

Total Energy of a System

Conservation of Mechanical Energy

$$(KE_t + KE_r + PE_g)_i = (KE_t + KE_r + PE_g)_f$$

other potential energies of any other conservative forces can be added.

Work-Energy in a Rotating System

For nonconservative force, the work-energy theorem:

$$W_{nc} = \Delta KE_t + \Delta KE_R + \Delta PE$$

Angular Momentum

Angular momentum is defined as

$$\vec{L} = I\vec{\omega} = m(\vec{r} \times \vec{v})$$

and

$$\sum \vec{\tau} = \frac{\Delta \vec{L}}{\Delta t}$$

It is similar to the relationship between force and momentum in a linear system.

If the net torque is zero, the angular momentum remains constant.

That is, when $\sum \vec{\tau} = 0$, $\vec{L}_i = \vec{L}_f$ or $I_i\vec{\omega}_i = I_f\vec{\omega}_f$.